

Facility: SALEM 1 & 2 Scenario No.: ESG-3 Op-Test No.: 13-01 NRC

Examiners: _____ Operators: _____

Initial Conditions: 100% power, MOL.

Turnover: Maintain current power. Perform weekly Vacuum Pump swap IAW S2.OP-SO.AR-0001.

Event No.	Malf. No.	Event Type*	Event Description
1		N PO	Swap condenser vacuum pumps
2	VL0336 VL0337	C CRS/PO	Loss of condenser vacuum (recoverable)
3	EL0144	C ALL	Loss of 2A 4160 volt vital bus (TS)
4	TU0055	I ALL	Turbine Inlet steam pressure transmitter (PT-505) failure (TS)
5	MS0302	M ALL	Steam dump failure with single MSIV failing to shut
6	RP0108	C CRS/RO	Failure of automatic Safety Injection (steamline D/P)
7	AF0181B AF0183	C ALL	Loss of All AFW flow – FRHS condensate pump recovery
			CT's: #1 Initiate manual Safety Injection #2 Establish Condensate flow in FRHS

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 13-01 NRC ESG-3

SCENARIO NUMBER: 13-01 NRC ESG-3

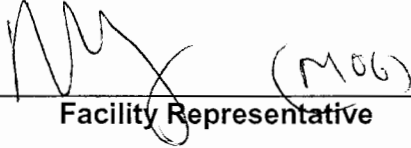
EFFECTIVE DATE: See Approval Dates below

EXPECTED DURATION: 75 minutes

REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary
New Issue for 13-01 NRC Exam

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>09-14-14</u> Date
APPROVED BY:	<u> J. J. Carney Operations Training Manager</u>	<u>10-23-14</u> Date
APPROVED BY:	<u> (MOG) Facility Representative</u>	<u>10-23-14</u> Date

I. OBJECTIVES

- A. Given the order or indications of a loss of vacuum, perform actions as the nuclear control operator to RESPOND to the malfunction, in accordance with S2.OP-AB.COND-0001.
- B. Given the order or indications of a loss of vacuum, DIRECT the response to the malfunction in accordance with S2.OP-AB.COND-0001
- C. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- D. Given the order or indications of a loss or malfunction of a safety related electrical distribution system, perform actions as the nuclear control operator to RESPOND to the loss or malfunction in accordance with S2.OP-AB.4KV-0001.
- E. Given indications of a loss or malfunction of a safety related electrical distribution system, DIRECT the response to the loss or malfunction in accordance with S2.OP-AB.4KV-0001.
- F. Given indication of unwarranted continuous rod motion perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-AB.ROD-0003.
- G. Given indication of unwarranted continuous rod motion DIRECT the response to the malfunction in accordance with S2.OP-AB.ROD-0003.
- H. Given the indication of excessive steam flow, perform actions as the nuclear control operator to RESPOND to excessive flow in accordance with S1/S2.OP-AB.STM-0001
- I. Given the order or indications of a reactor trip perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- J. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.
- K. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- L. Given a safety injection has occurred and equipment has failed to START equipment that has failed to automatically start in accordance with the approved station procedures.
- M. Given indication of excessive containment pressure DIRECT the response to the excessive pressure in accordance with the approved station procedures.
- R. Given the order or indications of excessive containment pressure perform actions as the nuclear control operator to RESPOND to the excessive pressure in accordance with the approved station procedures.
- S. Given the unit in the EOPs and an excessive containment pressure the crew will reduce containment pressure IAW approved station procedures.

II. MAJOR EVENTS

- A. Swap Condenser Vacuum pumps
- B. Loss of Condenser Vacuum (recoverable)
- C. Loss of 2A 4KV vital bus on Bus Differential
- D. Turbine Steamline Inlet Pressure Transmitter PT-505 failure
- E. Steam Dump failure
- F. Loss of all AFW / FRHS / Cond pump recovery

III. SCENARIO SUMMARY

- A. The crew will take the turnover at 100% power, MOL, with direction to swap operating condenser vacuum pumps per Control Room logs weekly pump swap using S2.OP-SO.AR-0001, Condenser Air Removal System Operation.
- B. During the vacuum pump swap, the pump being secured will have its air injection suction isolation valve AR25 fail open, causing condenser vacuum to degrade if not addressed immediately. The crew will attempt to re-start the affected pump (it will not start), direct field personnel to immediately shut the affected AR25, and start the remaining vacuum pump, which will restore vacuum.
- C. Once vacuum has been restored, the 2A 4KV vital bus will lock out on bus differential. The CRS will enter S2.OP-AB.4KV-0001, Loss of 2A 4160 Volt Vital Bus. The CRS will identify Tech Specs.
- D. The Main Turbine Steamline Inlet Pressure Transmitter (PT-505) will fail low. If control rods are in auto (they may have been placed in manual due to non-urgent failure alarm present) they will insert at maximum rate. The RO will verify no load reject in progress, and place control rods in manual. The CRS will enter S2.OP-AB.ROD-0003, Continuous Rod Motion. If control rods are in manual, the crew will diagnose the failure based on alarms received and RP4 hi steam flow bistable status.
- E. The crew will swap Main Steam Dumps if in S2.OP-AB.ROD-003, or IAW SO.RPS-0006 based on TSAS 3.3.2.1 requiring placing PT-505 in the tripped condition. If the crew did not require AB.ROD-3 entry and do not swap steam dumps to MS Pressure control (the TSAS gives them 6 hours to place channel in tripped condition) the steam dump failure will be entered by the Simulator Operator. The steam dump failure will cause 21-23TB20 steam dumps to open. The steam dumps will be unable to be shut or turned off. The CRS will direct a Rx trip and MSLI.
- F. The MSLI will only shut 3 of 4 MSIV's, and the CRS will direct a Safety Injection be initiated. An auto SI signal will be generated on Steamline Delta Pressure as 23 SG blows down. Auto SI initiation is failed.
- G. 22 AFW pump will trip shortly after AFW flow reduction is performed following TRIP-1 immediate actions. 23 AFW pump will trip 5 minutes later. The crew will isolate feedwater to 23 SG. The CRS will transition to FRHS-1 when directed in TRIP-1.
- H. The crew will perform a Condensate pump recovery in FRHS-1 due to SGFPs not being available, and all AFW pumps failed.
- I. The scenario will terminate when SG WR levels are rising in FRHS-1, or when RCS Bleed and Feed is initiated.

IV. INITIAL CONDITIONS

_____ Pre-snapped IC-233

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

<i>Initial</i>	Description
_____ 1	2VC1 and 2CV4 C/T
_____ 2	RCPs (SELF CHECK)
_____ 3	RTBs (SELF CHECK)
_____ 4	MS167s (SELF CHECK)
_____ 5	500 KV SWYD (SELF CHECK)
_____ 6	SGFP Trip (SELF CHECK)
_____ 7	23 CV PP (SELF CHECK)
_____ 8	Provide copy of S2.OP-SO.AR-0001 with prereq 2.2 (off normal check) signed off.
_____ 9	Provide blank copy of S2.OP-DL.ZZ-0003, Control Room logs Modes 1-4.
_____ 10	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	3	EVENT ACTION: kcf08dv1 // STM DUMP CONTROL MODE-MAIN STM P COMMAND: PURPOSE: <update as needed>
	5	EVENT ACTION: kb515tcp // 22AF21 CLOSE COMMAND: PURPOSE: <update as needed>
	7	EVENT ACTION: kck05anz // 25AR25 25 VACUUM PUMP-CONDENSER COMMAND: DMF VL0440 PURPOSE: <update as needed>
	9	EVENT ACTION: kck04tpz // 24 VACUUM PUMP-STOP COMMAND: PURPOSE: <update as needed>
	11	EVENT ACTION: kck02pbz // 22 VACUUM PUMP-START COMMAND: DMF CN0086B PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	VL0337 24AR25 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	100
02	EL0144 LOSS OF 2A 4160V VITAL BUS	N/A	N/A	N/A	RT-1	
03	TU0055 TURBINE INLET PRESS XMTR 505 FAILS H/	N/A	N/A	N/A	RT-3	0
04	RP0108 FAILURE OF AUTOMATIC SI	N/A	N/A	N/A	N/A	
05	AF0181B 22 AUX FEEDWATER PUMP TRIP	00:00:15	N/A	N/A	ET-5	
06	AF0183 23 AUX FW PMP OVERSPEED TRIP	00:01:00	N/A	N/A	ET-5	
07	VL0440 25AR25 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	0
08	CN0086B Loss of 22 main condenser vacuum	N/A	N/A	N/A	ET-9	4
09	MS0092G 23MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
10	VL0463 21TB20 Fails to Position (0-100%)	N/A	N/A	N/A	ET-3	100
11	VL0464 22TB20 Fails to Position (0-100%)	N/A	N/A	N/A	ET-3	100
12	VL0465 23TB20 Fails to Position (0-100%)	N/A	N/A	N/A	ET-3	100

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	CF06 F DI STM DUMP INTLK TRAIN 'A'-OFF	N/A	N/A	N/A	N/A	OFF
___ 02	CF07 F DI STM DUMP INTLK TRAIN 'B'-OFF	N/A	N/A	N/A	N/A	OFF
___ 03	CF10 F DI STM DUMP MANUAL DECREASE (CLOSE)	N/A	N/A	N/A	N/A	OFF
___ 04	CK04 E DI 24 VACUUM PUMP-START	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

Description

___ 1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Swap Condenser Vacuum Pumps When contacted, report Chemistry is ready for swapping vacuum pumps. Role play: When contacted, report			
	CRS directs PO to start the next sequential Condenser Vacuum pump (25), and secure the previous pump (24) IAW S2.OP-SO.AR-0001, Condenser Air Removal System Operation.		
	PO refers to section 5.5, Swapping Condenser Vacuum Pumps.		
	PO contacts Chemistry to inform them of vacuum pump swap.		
	PO determines section 5.5.4 for starting 25 vacuum pump and stopping 22 or 24 vacuum pumps is proper procedure section.		
	PO starts 25 vacuum pump.		
	PO contacts NEO to check 25 vacuum pump seal tank level, Seal Water Circulator discharge pressure is 10-25 psig as indicated by 2PL3226, and 25 Vacuum Pump suction pressure is at least 3" Hg. Above Main Condenser vacuum.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
you will check conditions and get back to them. One minute later, report 25 vacuum pump seal tank level is normal ½ way between high and low marks, Seal Water Circulator discharge pressure is 18 psig, and 25 vacuum pump suction pressure is 28" Hg.			
	PO opens 25AR25.		
	PO stops 24 vacuum pump.		
Simulator Operator: Ensure ET-9 is true when 24 vacuum pump stop PB is depressed. This inserts the loss of vacuum. 2. Loss of Main Condenser Vacuum			
	PO reports 24AR25 did not shut.		
	CRS directs start of 24 vacuum pump.		
	PO reports 24 vacuum pump will not start.		
	PO contacts NEO to isolate 24AR25 by closing 24AR23 .		
	PO reports slowly degrading condenser backpressure.		
Note: If CRS directs starting the remaining vacuum pump here, then			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
vacuum will improve, and S2.OP-AB.COND-0001 may not be entered.			
Simulator Operator: Ensure ET-11 is true when 22 vacuum pump start PB is depressed. This deletes the vacuum malfunction.			
Simulator Operator: 15 seconds after 22 vacuum pump is started, delete MALF VL0337, and if previously directed to locally shut 24AR23, report as NEO that 24AR23 is shut.			
	CRS enters S2.OP-AB.COND-0001, Loss of Condenser Vacuum.		
	CRS directs initiation of Attachment 1 CAS.		
	CRS ensures local operators are addressing vacuum problem.		
	PO starts all available vacuum pumps if not previously performed.		
	PO reports condenser backpressure is improving.		
	CRS goes to Section 4.0, Completion and Review to exit procedure.		
Continue to next event on direction			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
from Lead Evaluator.			
3. Loss of 2A 4160 volt bus			
Simulator Operator: Insert RT-1 on direction from Lead evaluator. MALF: EL0144 Loss of 2A 4160V Vital Bus.			
	RO/PO announce multitude of unexpected alarms, and diagnose the loss of 2A 4KV vital bus.		
	CRS enters S2.OP-AB.4KV-0001, Loss of 2A 4KV Vital Bus.		
	CRS directs initiation of Attachment 1 CAS.		
	PO reports 2A EDG running, and dispatches an operator to 2A EDG.		
Simulator Operator: If directed to deenergize 2A SEC, use Remote DG01D to YES.			
	RO reports 23 charging pump was in service, closes 2CV55, and starts 21 charging pump.		
	RO reports 23 charging pump stopped.		
	RO ensures RCP seal injection flow.		
	RO places 22 primary water pump in auto, and reports 22 BAT pump in auto.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports 21 RHR pump was not in service and 2A vital bus is deenergized.		
	PO reports no AFW pumps in service.		
	RO reports all SW and CCW low pressure alarms clear.		
	RO restores letdown by verifying 2CV2 and 2CV277 open and in auto, ensuring charging flow ~80 gpm, opening one 75 letdown orifice, and adjusting 2CV18 to maintain letdown pressure ~300 psig, then placing 2CV18 in automatic.		
	CRS dispatches operator and Maintenance to investigate loss of 2A bus.		
	CRS determines Tech Specs applicable to loss of 2A bus.		
	CRS directs initiation of S2.OP-AB.460-0001, Loss of 2A 460/230V Vital buses.		
	Applicable Tech Specs are: 3.9.12 for FHV – immediate 3.4.5 for 2PR6 loss of power – 1 hour 3.6.1.1 for 21SW20 being open w/no power – 1 hour 3.8.2.1 action a for 2A 4KV bus deenergized – 8 hours 3.8.2.3 and 3.8.2.5 for battery chargers – 2 hours 3.6.3 for CIV inoperable – 4 hours 3.7.1.2.a and b, action b for 2 AFW pumps –		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
AB.460-0001 steps start here – some action in this AB are redundant to actions taken in AB.4KV-1.	6 hours 3.7.6.a.1 CREACS fans – 4 hours 3.3.3.1 act. b Rad monitors		
	PO verifies centrifugal charging pp and letdown in service.		
	PO dispatches operators to swap to alternate battery chargers for 2A 125 and 28 VDC batteries.		
	PO starts 22 chilled water pump.		
	PO ensures alternate ventilation system components placed in service, and dispatches an operator to realign DFO to 22 DFO xfer pump to regular.		
Note: The rod control Non-urgent failure alarm will be in. The ARP states to place rod control in manual. This may or may not be identified when it is determined to proceed to next event based on the amount of alarms requiring verification. Continue to next event on direction from Lead Evaluator.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
4. Failure of Turbine Steamline Inlet Pressure Transmitter PT-505. Simulator Operator: Insert <u>RT-3</u> on direction from Lead Evaluator. MALF: TU0055 Turbine Inlet Press Xmtr fails low. Severity: 0 AB.ROD-3 steps here.			
	If rods are in auto, RO report insertion at 72 spm, no load reject, and places control rods in manual. If rods are in manual, RO reports OHA's and RP4 bistable indications.		
	Crew diagnoses PT-505 failed low.		
	If rods moved, CRS enters S2.OP-AB.ROD-0003, Continuous Rod Motion.		
	RO reports rods in manual and rod motion stopped , and rod motion was inward.		
	RO reports Tavg on program.		
	RO reports PT-505 failed low and rods above RIL.		
	PO aligns steam dumps by: - Ensuring steam pressure setpoint is 1005 psig. - Ensuring steam dumps in auto. - Depressing MS Pressure Control PB.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: Ensure ET-3 is true when steam dumps placed in MS Pressure Control. This opens all TB20 steam dump valves.			
AB-ROD-3 NOT entered steps start here.			
Note: There is no alarm response or abnormal procedure direction to swap steam dumps to MS Pressure control mode. The CRS can use S2.OP-SO.MS-0002, Steam Dump System Operation to swap steam dumps once it is identified that the steam dumps have a 100%Tavg demand.			
IF the crew does not identify that steam dump control mode should be swapped, then Lead Evaluator direct Simulator Operator to force ET-3 TRUE to initiate the steam dump failure.			
Note: All 3 TB20's failing open will result in total steam flow rising from 100% to 107%. An auto MSLI will not reach its initiation setpoint.			
	PO reports 21-23TB20's have opened.		
	CRS directs the PO to turn off steam dumps or		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Once the MSLI is initiated, 23 SG pressure will rapidly drop to more than 100 psig below the remaining SGs.	attempt closing dump valves in manual, and PO reports valves remain open.		
	CRS directs RO to trip the Rx and initiate a MSLI IAW CAS of S2.OP-AB.STM-0001.		
	RO trips the Rx.		
	RO initiates MSLI and reports 23MS167 remains open.		
	RO initiates MSLI on other SSPS train and reports 23MSIV remains open.		
	RO attempts to FAST close 23MS167 and reports it remains open.		
CT#1 (CT-2) Manually actuate SI prior to completion of TRIP-1 step 5.	RO reports an automatic SI signal is present on SG D/P between 23 SG and the remaining SG's, and reports SI has not actuated.		
	RO initiates SI and completes immediate actions by backing up main turbine trip and verifying at least one 4KV vital bus is energized.		

Evaluator/Instructor Activity		Expected Plant/Student Response	SBT LOG	Comment
SAT _____ UNSAT _____				
		RO and CRS verify immediate actions complete.		
		PO isolates AFW flow to 23 SG.		
		CRS dispatches operator to locally close 23MS167 or isolate 21-23TB20's or both.		
Simulator Operator: Ensure ET-5 is true when 22AF21 is throttled. This trips 22 AFW pump, then trips 23 AFW pump after a 1 minute delay.				
		While throttling AFW flow, PO reports 22 AFW pump has tripped.		
		PO adjusts 21, 22, and 24AF11's (from TDAFW pump) to control AFW flow to intact SGs.		
		PO reports no available equipment failed to start on SEC initiation.		
		PO reports 21 AFW pump has no power, 22 AFW pump has tripped, and 23 AFW pump is in service.		
		PO reports when 23 AFW pump trips.		
		PO reports 21SW20 and 21SW223 have no indication on 2RP4. (no power)		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: If the CRS arrives at step 20 with any SG NR level > 9%, then even with no AFW flow, will NOT go to FRHS-1. CRS should continue in TRIP-1 until a transition to LOCS-1 is made, at which time CFSTs would become active and the CRS could transition to FRHS-1 on RED path when it occurs. IF a transition to LOCS-1 is made, steam flow will be initiated when MS10's are adjusted to stabilize RCS temp.			
	RO reports 21/22CA330's shut, and containment pressure normal.		
	RO reports no indications of high steam flow.		
	PO reports 2A 4KV vital bus deenergized.		
	RO reports control room ventilation status, and ensures proper swgr ventilation.		
	RO reports 2 CCW pumps in service.		
	RO reports ECCS injection flow status for current RCS pressure.		
	PO reports AFW flow and SG NR level status.		
	CRS transitions to FRHS-1 <22E4 lbm/hr AFW flow and all SG NR levels <9%.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports it was not operator action which required FRHS-1 entry.		
	RO reports RCS pressure > intact SGs and RCS Thots >350°F.		
	RO reports 21 and 22 charging pumps available.		
	CRS reads Bleed and Feed transition Continuous Action Steps.		
	PO ensures 21-24GB4 and 21-24SS94 are shut.		
	PO reports no AFW pumps running or can be started.		
	If not previously performed, crew sends operators to investigate cause of AFW pump trips.		
	RO stops all RCPs.		
	PO reports condensate system in operation, and SGFPs not available.		
	RO reports SI has occurred.		
	PO verifies SI valve alignment.		
	RO resets SI and Phase A isolation, and reports Phase B reset.		
	RO opens 21/22CA330s.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Crew should not select a SG for depressurization which will require initiation of RCS Bleed and Feed sooner than would otherwise be required.	PO ensures 2B and 2C SECs reset.		
	PO reports SECs for energized vital buses reset.		
	PO reports 230V control centers reset.		
	Crew selects 22 or 24 SG for depressurization, whichever has a lower SG level.		
	PO reports steam dumps are not available.		
	RO reports MSLI performed, with 23MS167 remaining open.		
	PO fully opens selected SG MS10 valve.		
	CRS dispatches operator to open selected SG BF40 valve.		
Simulator Operator: 3 minutes after being dispatched, use Remote BF06A (22BF40) or BF08A (24BF40) with a one minute ramp to open selected BF40.			
	PO opens selected SG BF13.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO ensures selected SG BF22 is released.		
	PO opens 21 and 22CN48s.		
	PO shuts 21/22CN32s.		
	CRS waits at step 20.5 as selected SG depressurizes.		
CT#2 (CT-45) Establish feed flow into any SG before RCS Bleed and Feed is required. SAT _____ UNSAT _____	PO reports selected SG WR level is rising.		
Lead Evaluator terminate scenario upon transition back to procedure in effect, or after Bleed and Feed has been initiated.	PO reports no SG NR level is >9%, and that selected SG WR level is rising.		
	PO maintains selected SG pressure <575 psig.		
	RO reports Bleed and Feed have not been initiated.		
	CRS returns to procedure in effect, either TRIP-1 or LOSC-1.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0004, Power Operation
- F. S2.OP-DL.ZZ-0003, Control Room Logs Modes 1-4 (page 13)
- G. S2.OP-SO.AR-0001, Condenser Air Removal System Operation
- H. S2.OP-AB.COND-0001, Loss of Condenser Vacuum
- I. S2.OP-AB.ROD-0003, Continuous Rod Motion
- J. S2.OP-SO.MS-0002, Steam Dump System Operation
- K. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- L. 2-EOP-FRHS-1, Loss of Heat Sink

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 100 RCS BORON: 853 MWe 1220

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core Burnup: 5,000 EFPD

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

Swap to next sequential Condenser Vacuum pump after turnover IAW S2.OP-SO.AR-0001, Condenser Air Removal System Operation, as directed by Control Room Logs, S2.OP-DL.ZZ-0003, page 13.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

Heating steam is aligned to Unit 1.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
___ With Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct if applicable.
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 13-01 NRC-ESG-3 **REVIEWER:** P Williams

Initials	Qualitative Attributes
PW	1. The scenario has clearly stated objectives in the scenario.
PW	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
PW	3. The scenario consists mostly of related events.
PW	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
PW	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
PW	6. The events are valid with regard to physics and thermodynamics.
PW	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
PW	8. The simulator modeling is not altered.
PW	9. All crew competencies can be evaluated.
PW	10. The scenario has been validated.
PW	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
PW	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

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Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	8	Total malfunctions inserted: 5-8
GG	3	Malfunctions that occur after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	1	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	2	Crew Critical Tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

13-01 NRC ESG-3

CT#1 (CT-2) Manually actuate SI prior to completion of TRIP-1 step 5.

Basis: Failure to manually actuate SI under the postulated conditions constitutes misoperation or incorrect crew performance in which the crew does not prevent “degraded emergency core cooling system (ECCS)...capacity.” In this case, SI can be manually actuated from the control room. Therefore, failure to manually actuate SI also represents a failure by the crew to “demonstrate the following abilities:

- Effectively direct or manipulate engineered safety feature (ESF) controls that would prevent (degraded emergency core cooling system (ECCS)...capacity)
- Recognize a failure or an incorrect automatic actuation of an ESF system or component”
- Take one or more actions that would prevent a challenge to plant safety”

CT#2 (CT-45) Establish feed flow into any SG before RCS Bleed and Feed is required.

Basis: Failure to establish the minimum required feedwater flow rate, under the postulated plant conditions, results in “adverse consequences or significant degradation in the mitigative capability of the plant.” In this case, the minimum required feedwater flow rate can be established by performing the appropriate manual action.

Therefore,

failure to manually establish the minimum required feedwater flow rate also represents a failure

of the crew to demonstrate the ability to:

- Effectively direct or manipulate engineered safety feature (ESF) controls (assuming
- AFW) that would prevent a significant reduction of safety margin beyond that irreparably
- introduced by the scenario
- Recognize a failure or an incorrect automatic actuation of an ESF system or component
- (assuming AFW)
- Take one or more actions that would prevent a challenge to plant safety”

Finally, failure to manually establish feedwater flow rate under the postulated conditions is a “violation of the facility license condition.”

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>
<u>N</u>	TRANSIENTS with PCS Unavailable
<u>N</u>	Steam Generator Tube Rupture
<u>N</u>	Loss of Offsite Power
<u>N</u>	Loss of Switchgear and Pen Area Ventilation
<u>N</u>	LOCA

<u>Y/N</u>	<u>Event</u>
<u>N</u>	Loss of Service Water
<u>N</u>	Loss of CCW
<u>N</u>	Loss of Control Air
<u>N</u>	Station Black Out

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Containment Sump Strainers
<u>N</u>	SSWS Valves to Turbine Generator Area
<u>N</u>	RHR Suction Line valves from Hot Leg
<u>N</u>	CVCS Letdown line Control and Isolation Valves

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Gas Turbine
<u>Y</u>	Any Diesel Generator
<u>Y</u>	Auxiliary Feed Pump
<u>N</u>	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
<u>N</u>	Restore AC power during SBO
<u>N</u>	Connect to gas turbine
<u>N</u>	Trip Reactor and RCPs after loss of component cooling system
<u>N</u>	Re-align RHR system for re-circulation
<u>N</u>	Un-isolate the available CCW Heat Exchanger
<u>N</u>	Isolate the CVCS letdown path and transfer charging suction to RWST
<u>N</u>	Cooldown the RCS and depressurize the system
<u>N</u>	Isolate the affected Steam Generator that has the tube rupture(s)
<u>N</u>	Early depressurize the RCS
<u>N</u>	Initiate feed and bleed

Complete this evaluation form for each ESG.